

Summarising a familiar paper with local AI

Time: 1–2 hours

Your task is to find a good setup for accurately summarising a familiar paper's **Introduction** using local language models. Choose the paper yourself, then work with a **provider model of your choice**, such as Copilot, ChatGPT, Claude or Gemini, **from the very start and throughout the assignment** to develop and improve your approach.

Ask, for example: “What local setup and models could work on my computer?”, “What could be a good prompt for model X to summarize a paper?” and “How can I automate my experiment log?”

You may use the provider online for advice and troubleshooting. Once set up, your summarisation workflow must run **entirely on your own computer, using local models, without an internet connection or cloud model calls**.

What to do

1. **Choose a paper you know well.** This can be your own paper or another article you are very familiar with. Choose one whose content you can quickly check for accuracy. Use material you are permitted to process with your chosen tools.
2. **Build and test your local setup.** With your provider's help, choose a local model, provide the Introduction and generate a summary. You can use to use the local model via Ollama, which we introduced in the first session, and use the model via the PowerShell, or rather graphical interface, or an environment you build yourself if you are adventurous ;) Choose whichever approach works for you.
3. **Work with your provider model to improve it.** After your first test, discuss the result and ask for help choosing local models, refining prompts, adjusting model settings (e.g. temperature). If you encounter any problem, including something unexpected, try to solve it with your provider's help. You may use multiple local models, prompts or processing steps; maybe one local model performs better than another. Test the suggestions locally and check that your final workflow can run offline.
4. **Check and refine.** Compare each summary with the original Introduction. Does it accurately capture the background, knowledge gap and research aim? Does it preserve important qualifications? Identify missing points, misleading statements or invented information. Keep improving within the time available, then select your best version.

Keep a simple log

Start logging with your first experiment. Ask your provider to help automate this if useful. Briefly record:

- **Prompt:** the prompt or prompts you used.
- **Model:** the local model or models you used.
- **Settings or workflow:** the settings and steps needed to repeat the attempt, including which text you provided.
- **Output:** the resulting summary.
- **Your assessment:** what you think of the result, what worked or failed, and what you learned.

Any clear format is fine. You could also write this manually in a simple .docx or .txt file.

Optional extension

If the Introduction summary is accurate enough, try including the Methods, Results or the whole article. Check whether your approach still works as the input becomes longer and more complex, as you may need to increase context length. You can also try to ask different questions (‘what would be a good follow-up study’, ‘what is a strong/weak point of this paper’, etc.).

Bring to the session

Bring **only your best summary and the log for that version**. Include the settings or complete workflow needed to reproduce it. Be ready to explain what you did, what you learned and why you consider this your best version.

You do not need to bring earlier attempts. A perfect summary is not required; but be prepared to explain the remaining limitations you found.